

What's Inside an SSD?

An SSD's performance and reliability come from four essential components working together.

NAND Flash Memory

The NAND chip stores all your data, even when the power is off. Most modern SSDs use **3D TLC NAND** for the perfect balance of speed, durability, and value, while **QLC NAND** offers higher storage capacity at a lower cost.

SSD Controller

The controller is the **brain of the SSD**. It manages data flow, improves performance, corrects errors, and extends the drive's lifespan. A powerful controller ensures faster loading, smoother multitasking, and consistent performance.

PCB (Printed Circuit Board)

The PCB connects all SSD components together. High-quality, multi-layer PCBs provide better signal stability, efficient heat management, and long-term reliability. Most modern NVMe SSDs use the **M.2 2280** form factor for maximum compatibility.

DRAM Cache

DRAM acts as a high-speed memory buffer, helping the SSD quickly locate and manage files. SSDs with dedicated DRAM deliver faster performance during heavy workloads, while DRAM-less SSDs use **Host Memory Buffer (HMB)** technology for efficient everyday performance.

These components work together to provide **faster speeds, better durability, lower latency, and a smoother computing experience.**

